

Hypoglycemic and Antioxidant Effects of a Polyherbal Formulation in Alloxan Induced Diabetic Rats

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Abstract

Objective: The present study was conducted to determine the hypoglycemic and antioxidant potential of polyherbal formulation in alloxan induced diabetic rats.

Study Design: In five different groups an aqueous suspension of herbal mixture containing equal amounts of– *Azadirachta indica*, *Gymnema sylvestre*, *Momordica charantia*, *Syzigium cumini*, *Trigonella foenum* was administered orally once a day at a dose of 500 mg/kg body weight for 4 weeks. In these groups lipid peroxidation, reduced glutathione, glutathione *S*-transferase was estimated in brain and kidney and compared statistically.

Results: There was a significant increase of 27% and 80% in LPO in the brain and kidney respectively, in diabetic animals. The herbal and standard drug treatment decrease LPO level up to control group. GSH level decreased by 34% and 48% in brain and kidney of diabetic group which was restored after herbal and glibenclamide treatment. There was no significant change in GST level in diabetic and herbal treated groups in brain while diabetic group showed reduced GST level in kidney which was raised to normal level with herbal as well as standard drug treatment.

Conclusion: Polyherbal formulation appeared to be safe and effective in diabetes and also have antioxidant potential.

Keywords: *Azadirachta indica*, *Gymnema sylvestre*, *Momordica charantia*, *Syzigium cumini*, *Trigonella foenum*, *Diabetes mellitus*, antioxidants

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Introduction

Diabetes mellitus (DM) is characterized by hyperglycemia associated with impairment in insulin secretion and/or insulin action as well as alteration in intermediary metabolism of carbohydrates, proteins and

lipids. In future annual incidence rate of DM will increase worldwide, especially in India. It has been proposed that approximately 57 million Indians will be affected by DM by the year 2025¹.

Polyherbal drugs are considered to be more effective for the management of diabetes with Ayurvedic medicines. Diabetes mellitus represents a syndrome of metabolic disorder and complex pathophysiological interactions between hyperglycemia, insulin resistance and dysfunction of the β cells of pancreas. The available antidiabetic measures such as oral hypoglycemic agents, do not effectively control the delayed diabetic complications like nephropathy, neuropathy, retinopathy and cardiovascular diseases².

WHO gives special emphasis to herbal antidiabetic drug development with proper screening³. Synthetic hypoglycemic agents can produce serious side effects including hematological, hypoglycemia and disturbances in liver, brain, and kidney. In addition, they are not suitable during pregnancy⁴. Compared to synthetic drugs, herbal preparations are frequently considered to be safe because of least toxic and fewer side effects. Different studies were performed clinically and experimentally. Leaves and seeds of *Azadirachta indica*⁵, leaves of *Gymnema sylvestre*⁶, fruits of *Momordica charantia*⁷, seeds of *Syzigium cumini*⁸ and seeds of *Trigonella foenum*⁹ have been investigated. There are only few reports on polyherbal formulation when used in diabetic rats in low doses, therefore the present study was

planned to assess the antioxidant potential and hypoglycemic effect of the polyherbal formulation in low doses in diabetic rats, because different ingredients work via different mechanism for the same effect. In present study the doses of five herbal drugs (100 mg/kg each) used in the herbal formulation were significantly less than the individual dose used alone. The results of the present study can serve as a step toward the development of an antidiabetic polyherbal drug for diabetes.

Study design

Preparation of polyherbal formulation

Polyherbal formulation “*Momordica charantia* (Karela-fruit), *Azadirachta indica* (Neem-leaves), *Syzigium cumini* (Jamun-seeds), *Gymnema sylvestre* (Gurmar-leaves) and *Trigonella foenum* (Methi-seeds)” were purchased from the market. The plants were botanically authenticated from the department and powdered separately. Fresh herbal suspension (5%) was made in cold distilled water before use by mixing the five ingredients in equal ratio (w/w) and homogenized with the help of a pestle and mortar (till powder dissolved completely).

Experimental design

Ethical clearance for experiment was taken from institutional ethical committee. Diabetic rats having fasting blood glucose in the range of >200 mg/dl were selected for the study. Rats of either sex were

divided into the five equally sized groups (8 rats) for 4 weeks of study. These groups were: Group I-Control (Non diabetic & without any treatment), Group II - Diabetic (without herbal treatment), Group III - Herbal treated-control (Non- diabetic rats given polyherbal formulation), Group IV – herbal treated diabetic (Diabetic rats given polyherbal formulation), Group V - Diabetic treated-*Glibenclamide* (Diabetic rats given *Glibenclamide* treatment).

Induction of diabetes

Albino rats (100-150 g) maintained under standard laboratory conditions, were fasted overnight prior to alloxan administration. Fasting blood glucose of all animals was determined and found in the range of 75 to 90 mg%. Single dose of alloxan (150 mg/kg) dissolved in normal saline was administered Intraperitoneal (i.p.) to develop diabetes. Control animals were treated with normal saline. Glucose was determined again on fourth day of alloxan administration and the animals exhibiting fasting blood glucose range >200 mg/dl were used in the experiment. Blood glucose level was again estimated in all the groups at the end of study (4 weeks) before sacrificing the rats.

Administration of Polyherbal formulation

Freshly prepared homogenate of herbal formulation was administered orally to rats once daily with the help of a feeding

cannula at a dose of 500 mg/kg for 4 weeks starting from fifth day of alloxan administration. Non-diabetic group of rats also received polyherbal formulation similarly.

Sample collection

At the end of fourth week blood was collected by heart puncture in an EDTA vial (2 mg EDTA/ml blood) for glucose estimation. The rats were sacrificed by decapitation and brain & kidney tissues were taken out by dissection. Tissues were washed with ice-cold KCl-Tris buffer (1.15% KCl buffered with Tris-HCl, 0.01M, pH 7.4) and homogenized with the help of Potter-Elvehjem homogenizer in ice-cold KCl-Tris buffer containing 0.001M EDTA for GSH estimation and without EDTA for estimation of lipid peroxide and GST.

Determination of blood glucose¹⁰

Fasting blood glucose level of animals was estimated by glucometer (Accuchek Instant, Germany) using a drop of blood taken from the tail vein. At the end of polyherbal treatment blood collected in EDTA vial and blood glucose was determined spectrophotometrically by glucose oxidase method using commercially available kit (Accurex Biochemical Pvt. Ltd. India).

Determination of reduced glutathione and lipid peroxidation¹¹

The level of GSH in the homogenates of liver and kidney was estimated as protein free sulfhydryl content using Ellman's reagent. Malonyldialdehyde (MDA), a product of unsaturated fatty acid peroxidation, was estimated in tissue homogenate as a measure of thiobarbituric acid reactive substances (TBARS) formed from LPO.

Estimation of glutathione S-transferase

12

The activity of glutathione S-transferase (GST) in supernatant of tissues was determined using 1-chloro 2, 4-dinitrobenzine (CDNB) as the substrate.

3. Statistical analysis

The data was statistically analyzed by using ANOVA using Newman –Keuls test.

Results

We evaluate the antidiabetic and antioxidant potential of polyherbal formulation containing *Azadirachta indica*, *Gymnema sylvestre*, *Momordica charantia*, *Syzygium cumini* & *Trigonella foenum* (Table-1). Each of these plants has been reported to have antidiabetic and antioxidant activity.

An increase of 2-3 folds in the fasting blood glucose level was observed 5 days after alloxan administration, which persisted for another 4 weeks of the study period. In herbal and *glibenclamide* treated groups blood glucose level revert to normal value. Polyherbal formulation did

not elicit any significant change in the glucose level in the control group of rats (Figure 1). . There was a significant increase of LPO in diabetic group in kidney while herbal and standard drug treatment decrease LPO level up to control group (Figure 2). GSH level decreased in brain and kidney of diabetic group which was restored after herbal and glibenclamide treatment (Figure 3). There was no significant change in GST level in diabetic and herbal treated groups. In kidney diabetic treated group showed reduced GST level which was raised to normal level with herbal as well as standard drug treatment (Figure 4).

Discussion

Many medicinal plants have been traditionally used in India and other parts of the world, since long time for their sugar lowering effect. Herbal products are generally considered to be least toxic and free from side effects when compared with their synthetic counterparts for e.g. Polyherbal formulation - Diashis (*Syzygium cumini*, *Gymnema sylvestre*, *Holarrhena antidysenterica*, *Tinospora cordifolia*, *Pongamia pinnata*, *Asphultum*, *Psoralea corylifolia*, and *Momordica charantia*)' did not showed any metabolic toxicity¹³.

Glyoherb (Each 100 g of "Glyoherb"-sugar control granules (GH) contains: Gudmar Ext.: 15 gm; Mahamejva Ext.: 6 gm; Katuki Ext.: 6 gm; Chirata Ext.: 6 gm;

Karela Ext.: 6 gm; Indrajav Ext.: 3 gm; Amla Ext.: 6 gm; Gokshur Ext.: 3 gm; Harde Ext.: 6 gm; Jambubij Ext.: 6 gm; Methi Ext.: 3 gm; Neem patti Ext.: 6 gm; Chanraprabha: 3 gm; Arogyavardhini: 3 gm; Harida Ext.: 3 gm; Bang bhasma: 3 gm; Devdar Ext.: 3 gm; as the major active constituents) is claimed to be a single unique formulation that provides a holistic management of blood glucose- and diabetes-related complications. Administration of glucose (1.5 g/kg, PO) did not produce any significant change in the serum glucose levels of either control (90.15 ± 11.46 mg/dl) or only Glyoherb-treated group animals (70.91 ± 8.47 mg/dl) even at the end of 2 h, however, oral glucose administration, to diabetic control group animals markedly increased their glucose levels (506.58 ± 38.89 mg/dl). Serum glucose levels of diabetic rats treated with Glyoherb suspension in the dose of 200, 400 and 600 mg/kg or Glibenclamide were significantly decreased (258.05 ± 16.56 , 157.47 ± 33.63 , 222.31 ± 23.67 and 160.55 ± 45.16 mg/dl respectively) at the end of 120 min when compared with the diabetic control¹⁴.

Our findings are in agreement with Ferber et al¹⁵ that GST level decreased in kidney of diabetic rats which was raised to normal level with herbal as well as standard drug treatment. Reduced GST activity in

diabetic rats could be due to decreased availability of GSH and /or loss of catalytic efficiency of the enzyme resulting from oxidation of sulfhydryl and other group by ROS¹⁶. Significant decrease in antioxidant enzymes like catalase (CAT), peroxidase (Px), and glutathione-S-transferase (GST) in diabetes may be due to low levels of insulin or due to high levels of advanced glycated end products¹⁷.

Another polyherbal drug Garlip composed of aqueous extract of six medicinal plants showed a significant reduction of blood glucose level and decrease in tissue lipids and lipid peroxide formation¹⁸. Treatment with Diasulin (composed of 10 medicinal plants) and Glibenclamide resulted in a significant reduction of blood glucose level. Diasulin also resulted a significant decrease in tissue lipids and lipid peroxide formation¹⁹.

Halim EM²⁰ administered water extract of dried powder (200 mg/kg body wt) of *A. Augusta* and *A. indica* respectively orally to alloxan induced diabetic rats once a day for 8 weeks. Aqueous extract decreased the formation of lipid peroxides and increased antioxidants (superoxide dismutase, catalase, and glutathione peroxidase and glutathione transferase) in erythrocytes. There was reduction in LPO as TBARS in liver and kidney. It also prevented decrease in body weight. Our

findings are agreement with the statement of Halim et al that when *Abroma Augusta* and *A. indica* given together as water extract had hypoglycaemic action and had better effect than given alone.

Hypoglycemic activity of the herbal formulation used in the present study may be due to presence of several compounds having hypoglycemic activity and their multiple sites of action. This polyherbal formulation has no general toxic effect as body weights remain similar to those in the control. In oxidative injury lipid peroxidation increased as an indicator of tissue injury. The present study accomplished that an increase in oxidative stress occurs in diabetes as an increased level of LPO and decreased level of antioxidants (GSH & GST).

This formulation did not exert any toxic effects in kidney and brain functions. It was rather found to be improving kidney and brain functions. It possesses potential antioxidant activity as it decreases lipid peroxidation and enhances antioxidant status in diabetic rats. The antidiabetic activity of formulation may be attributed to its antioxidant properties also. Thus, data from the present study indicate antidiabetic and antioxidant properties of a polyherbal formulation against alloxan-induced diabetes in albino rats.

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Table 1. Composition of ingredient(s) present in polyherbal formulation

Botanical name (Common name)	Family	Part used	Ingredients Used (mg)
<i>Azadirachta indica</i> Margosa(Neem)	Meliaceae,	leaves	100
<i>Gymnema sylvestre</i> (Gudmar)	Asclepiadaceae.	leaves	100
<i>Momordica charantia</i> (Bitter gourd)	Cucurbitaceae	fruits	100
<i>Syzigium cumini</i> (Black plum)	Myrtaceae	seeds	100
<i>Trigonella foenum</i> (Fenugreek)	Fabaceae)	seeds	100

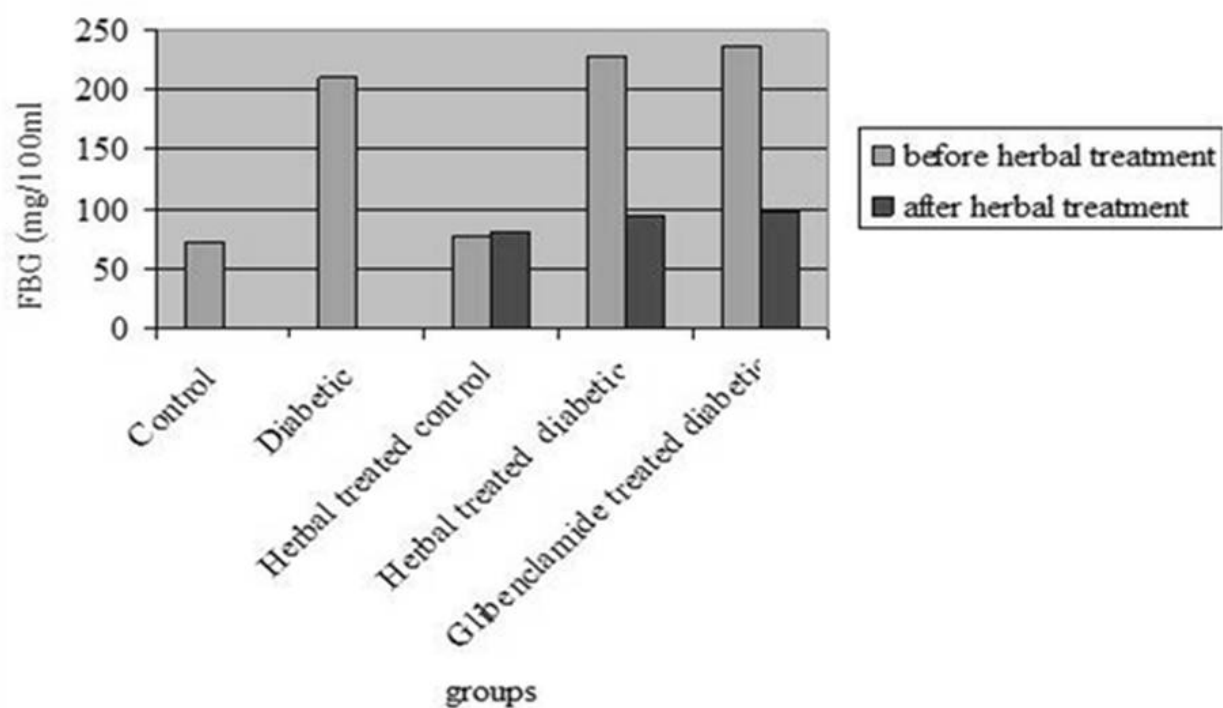
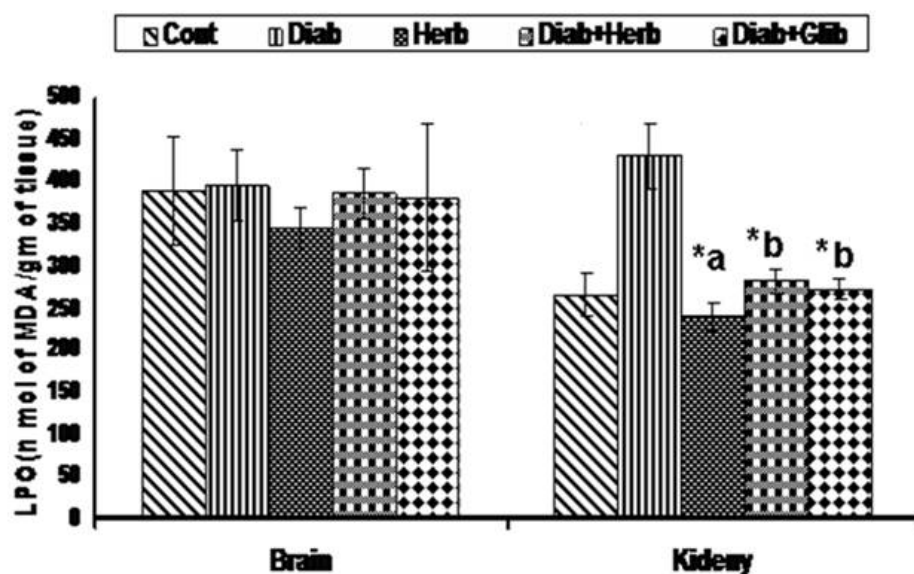
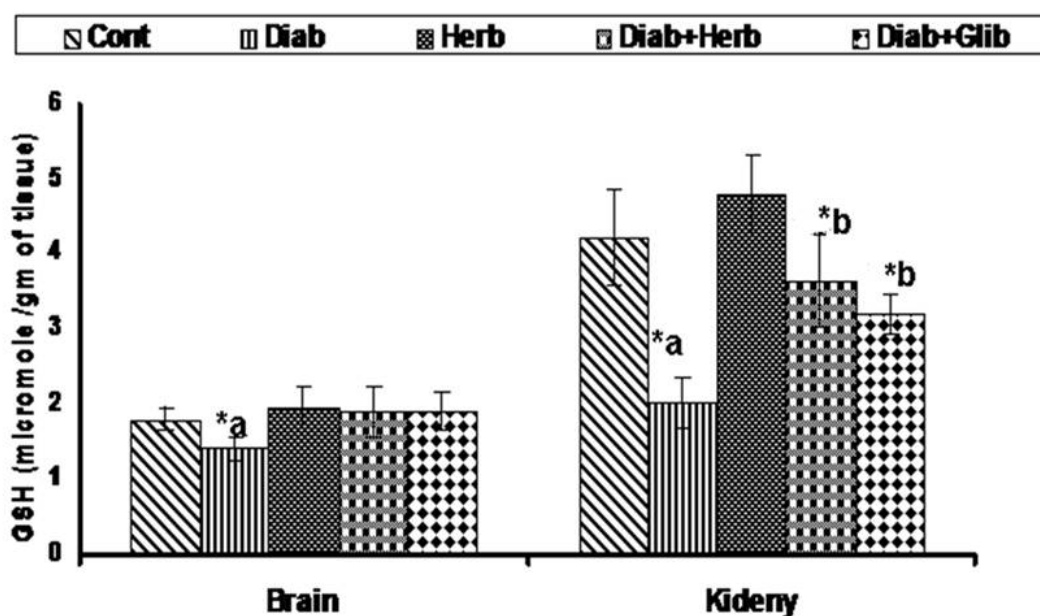


Fig 1. Diagram is showing blood glucose level in different groups



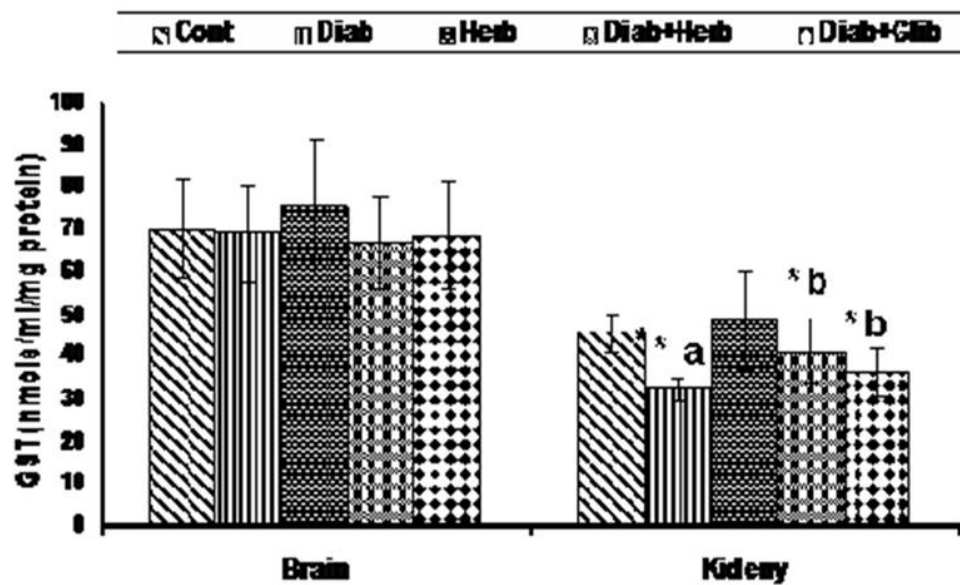
*a compared to control group. *b compared to diabetic treated group.

Fig 2. Diagram is showing LPO level in different groups



*a compared to control group. *b compared to diabetic treated group.

Fig 3. Diagram is showing GSH level in different groups



*a compared to control group. *b compared to diabetic treated group.

Fig 4. Diagram is showing GST level in different groups